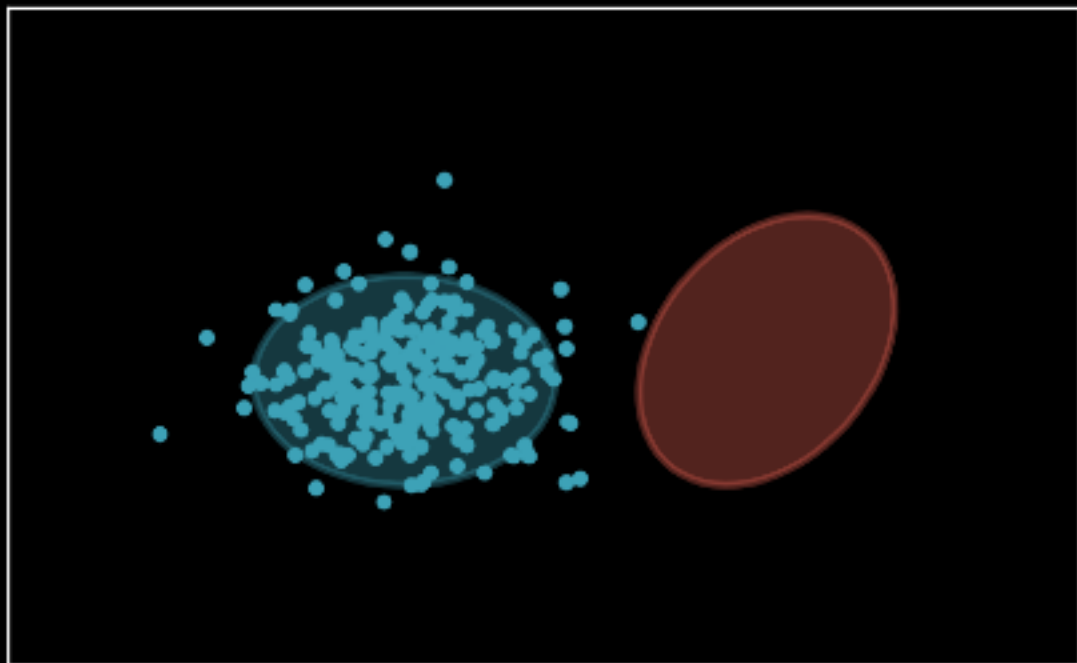


Schroedinger bridge between gaussians

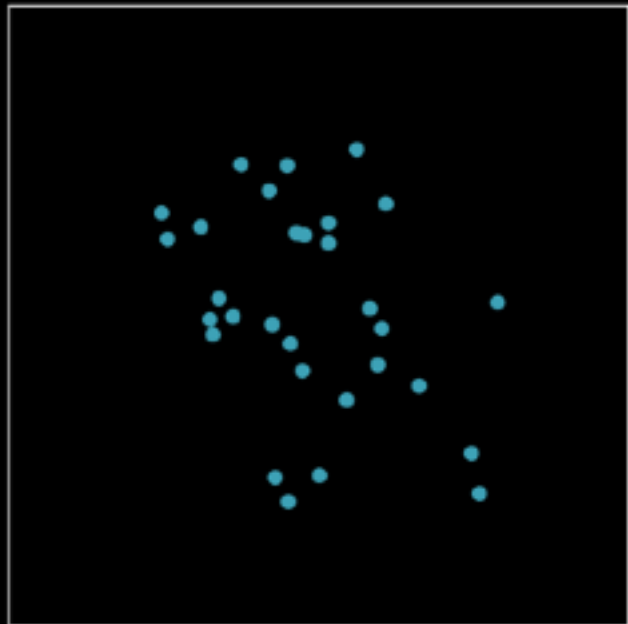




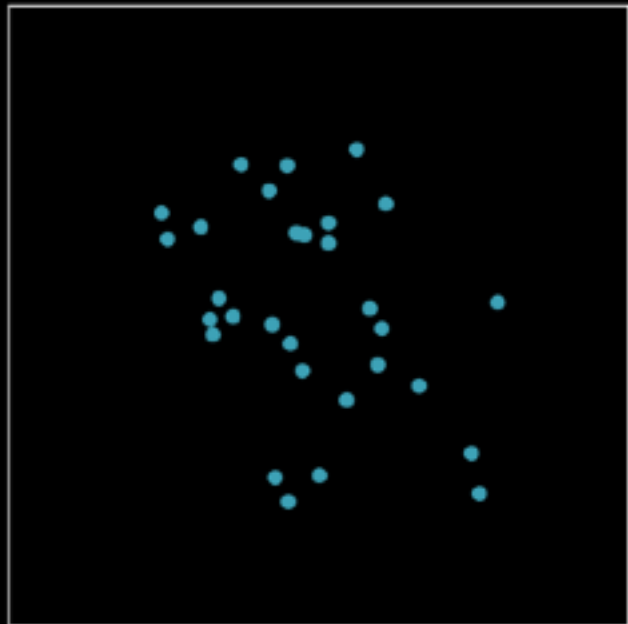
enforcing closeness to a reference diffuse path measure

most likely stochastic evolution

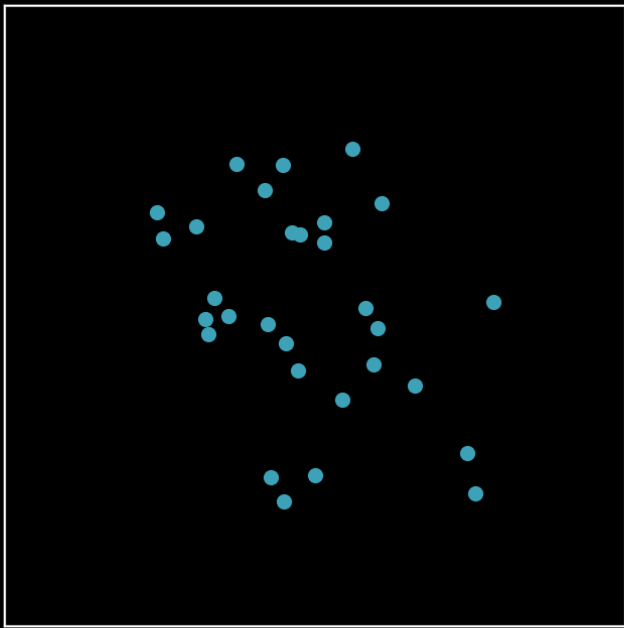
Brownian motion with trails ($k=30$)



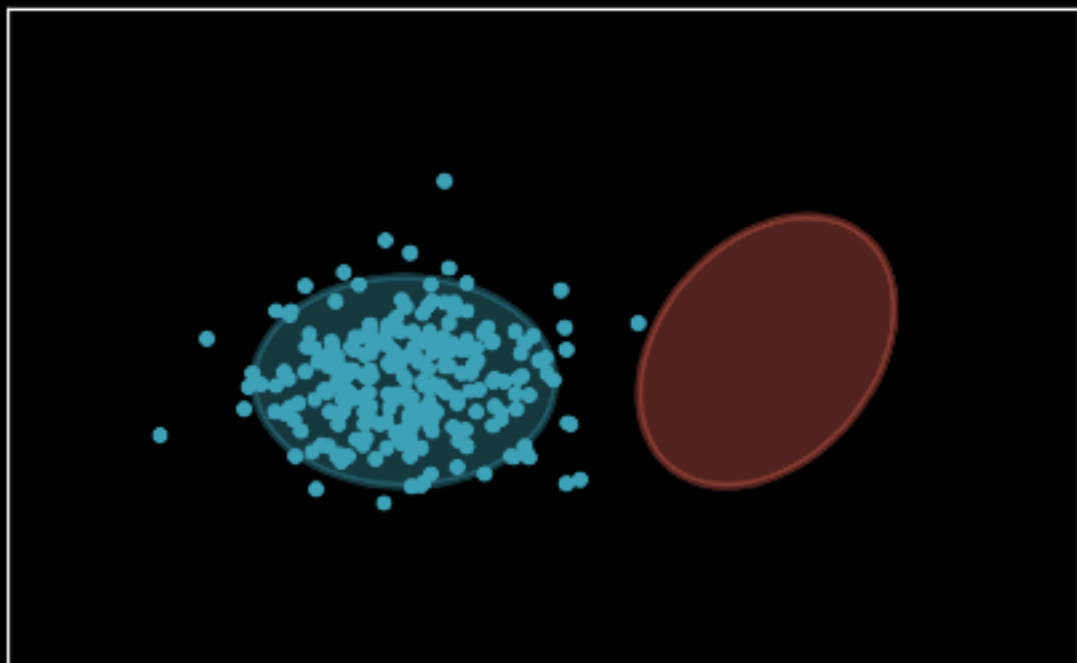
Brownian motion with trails ($k=30$)



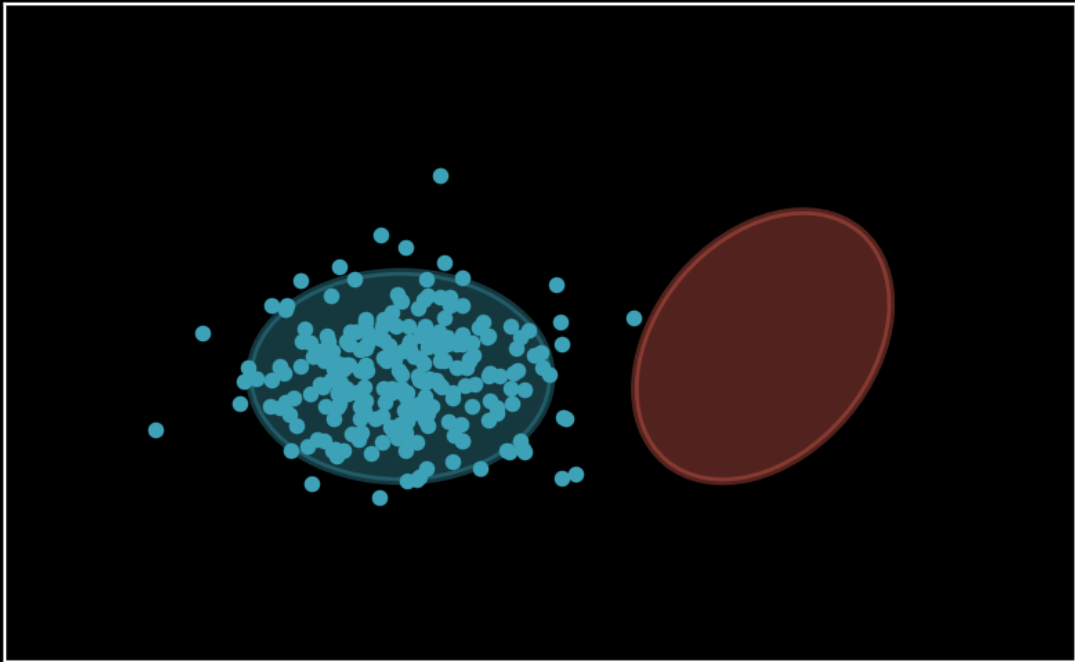
Brownian motion with trails (k=30)



Schroedinger bridge between gaussians

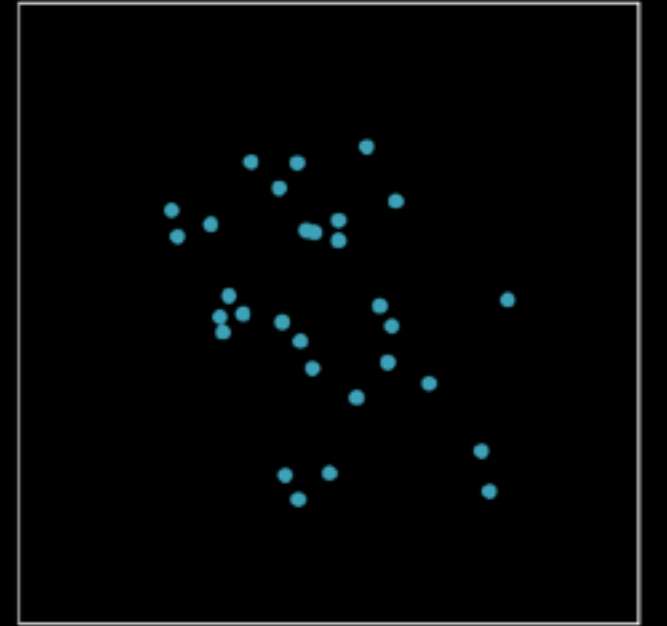


Schroedinger bridge between gaussians



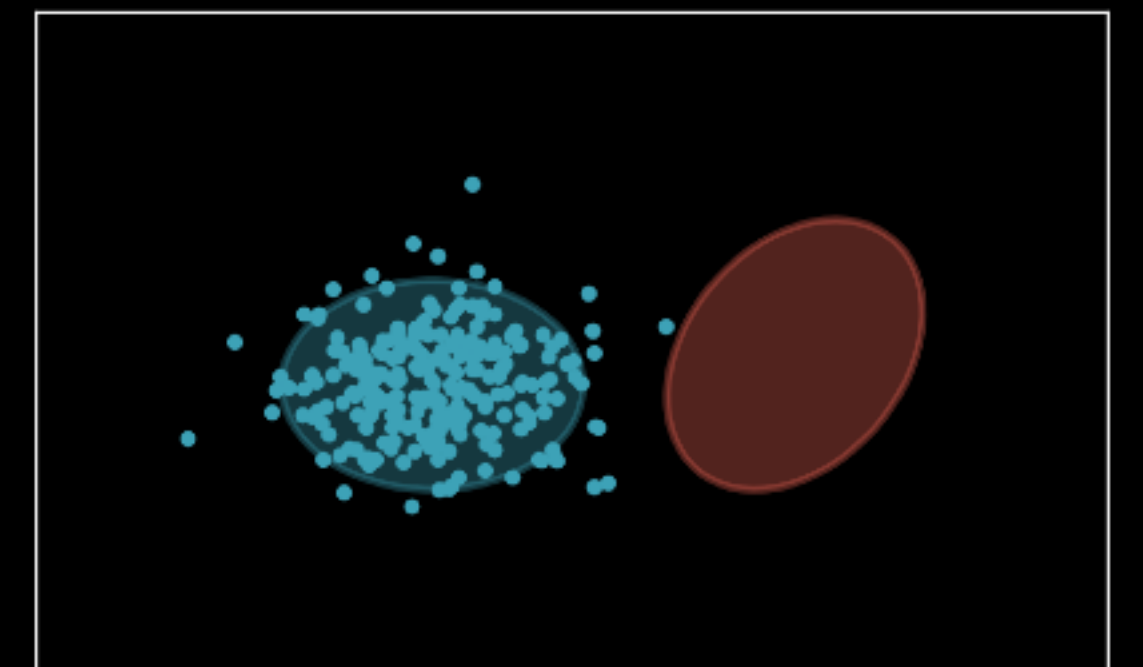
Schrödinger Problem

Brownian motion with trails (k=30)



- **Intuition:** Replace geodesics by noisy paths (Brownian motion).
- Regularize $\bar{\pi}$ by **enforcing closeness to a reference diffuse path measure**.
- Reference path measure: $\bar{\mathcal{K}}$ = law of reversible Brownian motion (formally, Wiener process).
- Schrödinger Bridge problem:
$$\min_{\bar{\pi} \in \mathcal{U}(\rho_0, \rho_1)} \text{KL}(\bar{\pi} \parallel \bar{\mathcal{K}})$$
- **Interpretation:** find the **most likely stochastic evolution** matching marginals.

Schroedinger bridge between gaussians



Schrödinger Problem